

PCCST501 - Computer Networks

Assignment 3: Data-Link Layer, Ethernet and Wireless LANs

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Q1. Data-link layer duties; LLC and MAC

Layer 2 is responsible for getting data from one device to the next device on the same link. Whatever the network layer hands down is put into a frame, sent across the cable or radio link, and checked on arrival.

- **Framing** - packs the packet between a header and a trailer so frame boundaries are clear.
- **Physical addressing** - MAC addresses say which NIC sent the frame and which should receive it.
- **Error checking** - a CRC value is attached; frames that fail the check are thrown away.
- **Flow control** - keeps the sender's speed in line with what the receiver can handle.
- **Access control** - sets the rules for taking turns on a shared link.

	LLC (IEEE 802.2)	MAC (802.3 / 802.11)
Where	Top half of layer 2, talks to layer 3	Bottom half of layer 2, talks to layer 1
What it does	Identifies the upper-layer protocol carried in the frame	Forms the frame, adds addresses and FCS, and controls access to the medium
Technology	Independent of the LAN type	Specific to each LAN technology
Example	Showing that the data is IPv4 or ARP	CSMA/CD in old Ethernet; CSMA/CA in Wi-Fi

Q2. Ethernet frame format

An Ethernet II frame is made of the fields in Figure 1 (sizes in bytes).

Preamble 7	SFD 1	Dest MAC 6	Src MAC 6	Type 2	Payload + pad 46-1500	FCS 4
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Figure 1: Ethernet II frame (sizes in bytes)

Field	What it is for
Preamble + SFD (7 + 1)	A bit pattern that synchronises the receiver and marks the point where the frame really starts.
Destination MAC (6)	Address of the receiving NIC - one device (unicast), a group (multicast) or everyone (FF:FF:FF:FF:FF:FF).
Source MAC (6)	Address of the sending NIC; this is what a switch records in its table.
Type / Length (2)	Names the protocol inside the frame (0x0800 = IPv4, 0x0806 = ARP) when the value is 0x0600 or higher; otherwise it gives the data length.
Payload (46-1500)	The packet from the network layer; padding is added if it is under 46 bytes.
FCS (4)	CRC-32 value used by the receiver to detect corruption; bad frames are discarded.

Minimum frame size is $14 + 46 + 4 = 64$ bytes and maximum is $14 + 1500 + 4 = 1518$ bytes (preamble and SFD not included).

Q3. Wireshark capture and MAC addresses of five frames

Steps. Open Wireshark and choose the interface in use. Create some traffic with `ping -c 5 8.8.8.8`, `arp -a` and a web page visit. After roughly 30 seconds stop the capture, enter the filter `arp || icmp || dns`, and open the *Ethernet II* section of each frame to note its source and destination MAC.

Note: Figure 2 is a documented example (sample capture), **not captured by me**, included as a reference as permitted by the course instructor. The MAC and IP addresses are illustrative.

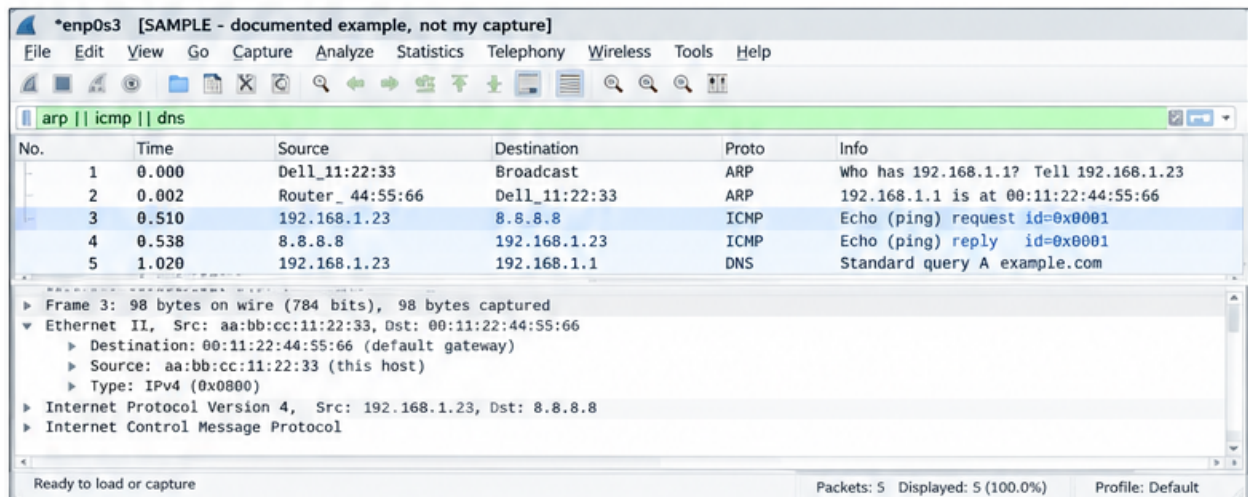


Figure 2: Wireshark sample capture showing five frames and MAC addresses (documented example, not my own capture)

Frame	Protocol	Source MAC	Destination MAC	EtherType
1	ARP request	aa:bb:cc:11:22:33	ff:ff:ff:ff:ff:ff	0x0806
2	ARP reply	00:11:22:44:55:66	aa:bb:cc:11:22:33	0x0806
3	ICMP request	aa:bb:cc:11:22:33	00:11:22:44:55:66	0x0800
4	ICMP reply	00:11:22:44:55:66	aa:bb:cc:11:22:33	0x0800
5	DNS query	aa:bb:cc:11:22:33	00:11:22:44:55:66	0x0800

Observation. Frame 1 goes to the broadcast address because the host has to ask who owns 192.168.1.1; the answer in frame 2 comes back as unicast. In frames 3 to 5 the destination MAC is the gateway's, even though the IP destination is far away, because a MAC address is used only for the next hop on the local network.

Q4. How a switch learns MAC addresses

A switch starts with an empty MAC table and fills it by watching traffic. When a frame comes in, the switch notes the source MAC against the incoming port. It then looks at the destination: a known address is sent out of that one port; an unknown or broadcast address is sent out of every other port (flooding); and if the destination is on the port the frame came from, the frame is dropped. Example with hosts A, B, C, D on ports 1-4:

Step	Frame	Switch action	Table afterwards
1	A to B	Learns A on port 1; B is unknown, so floods to ports 2, 3, 4	A:1
2	B to A	Learns B on port 2; A is known, so sends only to port 1	A:1, B:2
3	C to A	Learns C on port 3; sends only to port 1	A:1, B:2, C:3
4	D to B	Learns D on port 4; sends only to port 2	A:1, B:2, C:3, D:4
5	A to C	Both known; sends only to port 3	unchanged

The learned table can be viewed on a Cisco switch with `show mac address-table`.

Q5. Hub, bridge and switch compared

Point	Hub	Bridge	Switch
OSI layer	Layer 1	Layer 2	Layer 2 (some models layer 3)
Forwarding	Sends every bit out of all other ports	Passes frames between segments after checking its MAC table	Sends each frame only to the port of its destination, using hardware
Collision domains	One for all ports	One for each segment	One for each port
Duplex	Half duplex	Half duplex	Full duplex possible
Bandwidth	Divided among all hosts	Divided within a segment	Full speed on every port
Practical use	No longer used	Mostly replaced by switches	Used in all modern LANs

Q6. CSMA/CD and why it matters less now

CSMA/CD managed access on old half-duplex Ethernet where many hosts shared one cable. A host waits until the cable is quiet, then sends while still listening. If it sees a collision it transmits a short jam signal, stops, and tries again after a random delay from binary exponential back-off (maximum 16 attempts). Frames are at least 64 bytes so the sender is still active when the collision news arrives.

Today it hardly matters because:

- each host plugs into its own switch port, so there is nothing to collide with;
- full-duplex links transmit and receive on separate wires, so CSMA/CD is simply disabled;
- 1 Gbps and faster Ethernet is used in full duplex only;
- it appears only with legacy hubs or when duplex settings do not match.

Q7. CSMA/CA in 802.11 versus CSMA/CD

A Wi-Fi radio cannot receive while it is sending, and two stations may be out of each other's range (hidden terminals). So 802.11 uses **CSMA/CA**, which prevents collisions rather than detecting them:

- Listen to the channel. If it stays idle for DIFS, transmit; if not, wait.
- Draw a random back-off number and count it down only during idle time.
- After the frame is sent, the receiver waits SIFS and returns an ACK. A missing ACK means the frame is resent with a bigger back-off range.
- RTS/CTS can be used so that hidden stations hear the CTS and delay their sending (NAV).

Point	CSMA/CD	CSMA/CA
Goal	Find collisions and recover	Prevent collisions
Used on	Wired Ethernet	Wireless LAN
ACK	Not used at MAC layer	Needed for unicast frames
Back-off	Applied only after a collision	Applied before sending and after a failure
Hidden terminal	No problem	Solved with RTS/CTS and NAV

Q8. WLAN design for a 12 m x 8 m laboratory

With about 40 workstations and extra laptops, around 60 devices may be online together, so the room is covered by three ceiling-mounted access points.

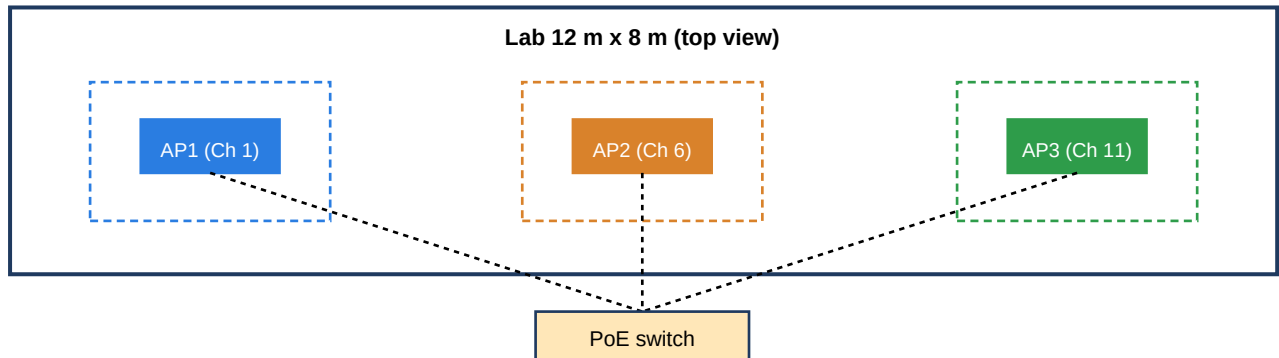


Figure 3: Proposed layout - three APs on channels 1, 6, 11 wired to a PoE switch

Item	Design
AP placement	Three APs fixed to the ceiling (about 2.7 m) in a row along the length of the room, cells overlapping by 15-20%, kept clear of metal cupboards.
Standard	Dual-band Wi-Fi 5/6, with 5 GHz preferred and 2.4 GHz for older devices.
Channels	2.4 GHz: channels 1, 6 and 11 for AP1, AP2, AP3. 5 GHz: 20/40 MHz channels like 36, 40, 44.
SSID	College-Lab for students and staff, College-Guest (isolated) for visitors; identical SSID on every AP for roaming.
Addressing	Lab network 192.168.10.0/24, gateway .1, DHCP range .50-.250. APs get static addresses in a management VLAN. Guest network 192.168.20.0/24.
Backbone	Gigabit PoE switch powers the APs and connects to a router/firewall with NAT.
Security	WPA3 (or WPA2-AES), 802.1X with RADIUS where available, WPS disabled, default passwords changed, guest isolation, regular firmware updates.
Capacity	Around 20 clients per AP; guest bandwidth capped.

Q9. Mobile IP and the need for it

An IP address both identifies a device and tells routers which network to deliver to. If a device moves and keeps its old address, packets still arrive at the old network and running TCP connections fail. **Mobile IP** (RFC 5944) solves this by letting a device keep a fixed home address wherever it goes.

Term	Role
Mobile node (MN)	The moving device.
Home agent (HA)	Router in the home network that keeps track of the MN and redirects its packets.
Foreign agent (FA)	Router in the visited network that supplies a care-of address.
Home / care-of address	Fixed address of the MN / temporary address tied to its present location.

- **Discovery:** HA and FA broadcast advertisements, so the MN can tell it is away from home.
- **Registration:** the MN reports its care-of address to the HA.
- **Tunnelling:** the HA intercepts packets sent to the home address and places them inside new IP packets (IP-in-IP) addressed to the care-of address, where they are unpacked.
- **Return traffic:** the MN answers directly, or via a reverse tunnel if ingress filtering would block it.

As a result, calls, downloads and remote logins continue when a phone or laptop switches between networks. Its main weakness is triangle routing, which Mobile IPv6 removes through route optimisation.

Q10. Wired Ethernet frame versus 802.11 frame

802.11 headers can be seen in Wireshark only if the Wi-Fi adapter is in monitor mode, so the standard frame layouts are compared here (documented layouts, not my own capture).

Preamble 7	SFD 1	Dest MAC 6	Src MAC 6	Type 2	Payload + pad 46-1500	FCS 4
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Figure 4a: Ethernet II frame

Frame Ctrl 2	Dur/ID 2	Addr1 6	Addr2 6	Addr3 6	Seq 2	Addr4 6 (opt)	Body 0-2312	FCS 4
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Figure 4b: IEEE 802.11 data frame (generic MAC header)

Feature	Ethernet II	802.11
Header size	14 bytes	24 bytes (30 with Addr4, 2 more with QoS)
Address fields	2	Up to 4, interpreted using the To-DS / From-DS bits
Frame kinds	Data only	Management, control (RTS, CTS, ACK) and data
Acknowledgement	None	Every unicast frame is acknowledged
Duration field	None	Used for virtual carrier sensing (NAV)
Sequence number	None	Sequence Control field detects duplicates and handles fragments
Maximum payload	1500 bytes	About 2304-2312 bytes
Security	None at layer 2	WPA2 CCMP / WPA3 GCMP, shown by the Protected bit
FCS	CRC-32	CRC-32

Conclusion. Radio is shared and unreliable, so 802.11 frames need more addresses, control flags, timing, sequencing, acknowledgements and encryption than a frame on a switched wired LAN.

References

- B. A. Forouzan, *Data Communications and Networking*, McGraw-Hill.
- A. S. Tanenbaum and D. Wetherall, *Computer Networks*, Pearson.
- IEEE 802.3 and IEEE 802.11 standards.
- RFC 5944, IP Mobility Support for IPv4, Revised.
- Wireshark User's Guide, <https://www.wireshark.org/docs/>